Sahaj Bhadja

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Summary

Software Engineer with 2+ years delivering **data-intensive applications**, **AI workflows**, and customer-focused solutions using Python, Java, TypeScript, SQL, cloud infrastructure, APIs, and modern frontend frameworks.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Object-Oriented Programming, Software Engineering, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, California*

- Architected a multi-tenant data platform using Azure SQL and Blob Storage with row-level isolation and normalized schemas for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Built data pipelines across FastAPI, PostgreSQL, SQL Server, and cloud storage, processing 5K+ monthly jobs with incremental loads, reconciliation, error handling, and CI/CD releases achieving 99%+ reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, California*

- Engineered an AI grading application using multimodal models and OCR to evaluate 10K+ daily text and handwritten submissions, delivering 90%+ grading consistency and reducing user feedback turnaround by 60%.
- Designed concurrent Python processing and secure multi-format ingestion through REST APIs and AWS RDS, executing 1K+ files per batch with fault-tolerant recovery, usage tracking, and reliable production support.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer, SecureSync *India*

- Designed object-oriented API integrations and AWS Glue transformations for JSON, CSV, and Parquet data, improving response times 30% and pipeline efficiency 68% while reducing technical debt 20%.
- Automated nightly builds, installations, integration tests, unit tests, and reports through Jenkins, validating reliable cross-platform synchronization while collaborating directly with Microsoft and GitHub stakeholders during Agile delivery.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern, VibrantChem *India*

- Implemented eight modular microservices for authentication, permissions, billing, resources, customers, leads, and notifications, reliably synchronizing 2M+ records through REST APIs and event-driven WebSocket updates across production workflows.
- Built responsive React workflows and automated Docker deployments through Jenkins CI/CD, achieving 95%+ build validation while integrating operational monitoring and analytics across controlled staging and production environments.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Developed Flask APIs, payment webhooks, pricing, and booking workflows while synchronizing Firebase events and processing hourly Databricks micro-batches across 50K+ operational records with consistently reliable sub-200-millisecond responses.
- Designed end-to-end workflows, mapped 12+ integration points, and presented technical solutions to client stakeholders, reducing design errors 25% while supporting effective collaborative cross-functional implementation planning and alignment.

Skills

Languages & Frontend: Python, Java, C++, TypeScript, JavaScript, SQL, HTML, CSS, React, Next.js, Angular, TailwindCSS
Data & Backend: Data Pipelines, ETL, FastAPI, Flask, Node.js, REST APIs, Microservices, WebSockets, PostgreSQL, SQL Server, MySQL, MongoDB, Firebase
Cloud & Delivery: AWS, Azure, GCP, S3, Blob Storage, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Git, Linux
Engineering: System Design, Data Structures, Algorithms, OOP, Unit Testing, Integration Testing, Debugging, Monitoring, Agile/Scrum, Requirements Analysis, Stakeholder Communication

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Built an AI-powered SQL development workspace supporting 10+ databases, accelerating query authoring, schema exploration, and data management through a unified cross-platform interface spanning Java and TypeScript services.

Big-Brain Live | *LLaMA-2, FAISS, PostgreSQL, GCP*

- Built secure document ingestion and collaborative AI workflows for 200+ users using Clerk, PostgreSQL, LLaMA-2, and FAISS, sustaining 95% query accuracy while reducing document review time by 60%.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*

- Built a data-intensive fintech application supporting 50K+ users and 1M+ transactions with secure third-party integrations, real-time synchronization, advanced filtering, and reliable Kubernetes-based deployment workflows at production scale.